SCENARIO-BASED REQUIREMENT ANALYSIS

Question:

Online Course Learning Management System: Develop an e-learning platform where instructors can upload courses and learning materials while students can enroll, attend assessments, track progress, and receive certificates.

1. Study and Analyze the Scenario

1.1 Business Domain

The proposed system belongs to the **Education and E-Learning domain**. The Online Course Learning Management System (LMS) provides a digital platform for instructors to create and manage courses and for students to learn, attend assessments, track their learning progress, and receive certificates after successful completion.

1.2 Problem Description

Traditional learning methods require students to attend physical classes and depend on printed learning materials. Students may also face difficulties in tracking their course progress, attending assessments, and obtaining course completion certificates.

Instructors need a centralized platform to upload course content, manage students, conduct assessments, and monitor student performance.

Therefore, an Online Course Learning Management System is proposed to provide a centralized, accessible, and user-friendly platform for online education.

1.3 Stakeholders

The major stakeholders of the system are:

1. **Administrator**
    - Manages users and system activities.
    - Monitors courses and platform operations.
2. **Instructor**
    - Creates and manages courses.
    - Uploads learning materials.
    - Creates assessments.
    - Monitors student performance.
3. **Student**
    - Registers and logs into the system.

- Browses and enrolls in courses.
- Accesses learning materials.
- Attends assessments.
- Tracks course progress.
- Receives certificates after successful completion.

1.4 Existing Challenges

- Difficulty in managing online courses through multiple platforms.
- Manual management of learning materials and student records.
- Lack of centralized progress tracking.
- Difficulty in conducting and managing assessments.
- Manual certificate generation.
- Limited monitoring of student performance.
- Students may not have a clear view of their course completion status.

1.5 System Objectives

The main objectives of the system are:

- To provide a centralized online learning platform.
- To allow instructors to create and manage courses.
- To allow instructors to upload videos, documents, and other learning materials.
- To enable students to browse and enroll in courses.
- To conduct online assessments.
- To track student learning progress.
- To evaluate student performance.
- To automatically generate certificates for eligible students.
- To provide secure and reliable access to the system.

2. Functional and Non-Functional Requirements

2.1 Functional Requirements

FR1 – User Registration

The system shall allow new students and instructors to create accounts using required personal information.

FR2 – User Login

The system shall authenticate registered users using their login credentials.

FR3 – User Profile Management

The system shall allow users to view and update their profile information.

FR4 – Course Creation

The system shall allow instructors to create new courses by providing course title, description, category, duration, and other required details.

FR5 – Course Management

The system shall allow instructors to edit, update, publish, and remove their courses.

FR6 – Learning Material Upload

The system shall allow instructors to upload learning materials such as:

- Videos
- PDF documents
- Presentations
- Notes
- Assignments
- Other educational resources

FR7 – Course Browsing

The system shall allow students to browse and search available courses based on categories, course names, or instructors.

FR8 – Course Enrollment

The system shall allow students to enroll in available courses.

FR9 – Learning Material Access

The system shall allow enrolled students to access the learning materials associated with their courses.

FR10 – Assessment Management

The system shall allow instructors to create and manage quizzes, tests, and assessments.

FR11 – Online Assessment

The system shall allow students to attend assessments online.

FR12 – Assessment Evaluation

The system shall automatically evaluate objective-type assessments and calculate student scores.

FR13 – Progress Tracking

The system shall track completed lessons, assessments, and overall course progress.

FR14 – Performance Monitoring

The system shall allow students and instructors to view assessment scores and learning performance.

FR15 – Certificate Generation

The system shall generate a course completion certificate when a student satisfies the required completion criteria.

FR16 – Certificate Access

The system shall allow students to view and download their certificates.

FR17 – Notifications

The system shall notify users about important activities such as course enrollment, new learning materials, assessments, results, and certificate availability.

FR18 – Administrator Management

The administrator shall be able to manage users, courses, and system activities.

2.2 Non-Functional Requirements

1. Performance

The system should respond to user requests within a reasonable time and should support multiple users simultaneously.

2. Security

The system shall protect user credentials and personal information through authentication and authorization mechanisms.

3. Reliability

The system should operate reliably and prevent loss of course, assessment, and progress data.

4. Usability

The system should provide a simple and user-friendly interface that can be easily understood by students and instructors.

5. Scalability

The system should be capable of supporting an increasing number of students, instructors, courses, and learning materials.

6. Maintainability

The system should be designed in a modular manner so that new features and modifications can be implemented easily.

7. Availability

The platform should be available to users whenever online learning services are required, except during scheduled maintenance.

8. Compatibility

The system should work properly on commonly used web browsers and different screen sizes.

3. Actors and Use Cases

3.1 Primary Actors

Administrator

Major use cases:

- Login
- Manage users
- Manage courses
- Monitor system
- Manage platform content

Instructor

Major use cases:

- Register/Login
- Create course
- Edit course
- Upload learning materials
- Create assessments
- View student performance
- Update course content

Student

Major use cases:

- Register/Login
- Browse courses
- Search courses
- Enroll in course
- Access learning materials
- Attend assessments

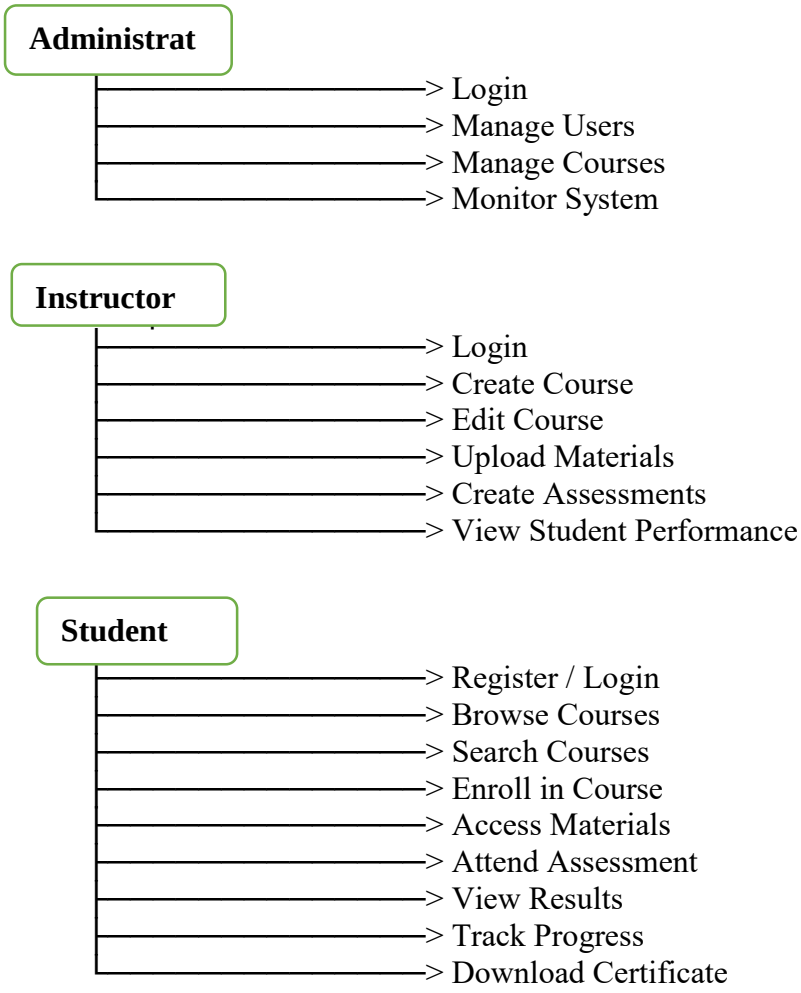
- View results
- Track progress
- Download certificate

3.2 Secondary Actors

- **Email/Notification Service** – sends notifications to users.
- **Database System** – stores user, course, assessment, progress, and certificate information.
- **File Storage System** – stores videos, documents, notes, and other learning materials.

3.3 Use Case Diagram

The major interactions between the actors and the Online Course Learning Management System are represented below.



4. Software Requirement Specification (SRS)

4.1 Introduction

The Online Course Learning Management System is a web-based e-learning platform designed to provide online education services. The system connects instructors and students through a centralized platform.

Instructors can create courses, upload learning materials, and conduct assessments. Students can enroll in courses, access learning resources, attend assessments, monitor their progress, and receive certificates after successful completion.

4.2 Problem Description

Students and instructors may face difficulties when learning resources, assessments, progress information, and certificates are managed separately.

The proposed system solves this problem by providing a single platform for course management, online learning, assessment, progress tracking, and certificate generation.

4.3 Scope of the System

The system covers:

- User registration and authentication.
- Instructor course management.
- Learning material management.
- Student course enrollment.
- Online assessments.
- Student progress tracking.
- Performance evaluation.
- Certificate generation.
- User notifications.
- Administrator management.

The system can be used by educational institutions, training organizations, online instructors, and independent learners.

4.4 Functional Requirements

The major functional requirements are:

1. User registration and login.
2. Profile management.
3. Course creation and management.
4. Learning material upload.
5. Course search and browsing.
6. Student enrollment.
7. Learning material access.
8. Assessment creation.

9. Online assessment participation.
10. Assessment evaluation.
11. Progress tracking.
12. Performance monitoring.
13. Certificate generation.
14. Certificate download.
15. Notifications.
16. Administrator management.

4.5 Non-Functional Requirements

The system shall satisfy the following quality requirements:

- High performance.
- Secure authentication.
- Reliable data storage.
- User-friendly interface.
- Scalability.
- Maintainability.
- Availability.
- Browser compatibility.
- Data privacy.

4.6 Use Case Descriptions

Use Case 1: Student Registration

Actor: Student

Precondition: Student does not have an existing account.

Main Flow:

1. Student opens the registration page.
2. Student enters required details.
3. System validates the information.
4. System creates the student account.
5. Registration is completed successfully.

Postcondition: Student can log into the system.

Use Case 2: Course Enrollment

Actor: Student

Precondition: Student must be logged into the system.

Main Flow:

1. Student browses available courses.

2. Student selects a course.
3. Student views course information.
4. Student selects the enrollment option.
5. System records the enrollment.
6. Student gets access to the course materials.

Postcondition: Student becomes an enrolled learner.

Use Case 3: Create Course

Actor: Instructor

Precondition: Instructor must be logged into the system.

Main Flow:

1. Instructor selects "Create Course".
2. Instructor enters course details.
3. Instructor uploads required materials.
4. Instructor saves the course.
5. System stores the course information.
6. Course is published or submitted for publication.

Postcondition: Course becomes available to students.

Use Case 4: Attend Assessment

Actor: Student

Precondition: Student must be enrolled in the course.

Main Flow:

1. Student opens the assessment.
2. System displays the questions.
3. Student submits answers.
4. System evaluates the answers.
5. System calculates the score.
6. Result is stored in the system.

Postcondition: Student's assessment result is recorded.

Use Case 5: Track Progress

Actor: Student

Precondition: Student must be enrolled in a course.

Main Flow:

1. Student opens the progress section.
2. System retrieves completed lessons and assessments.
3. System calculates course completion percentage.
4. System displays the progress to the student.

Postcondition: Student can view current course progress.

Use Case 6: Generate Certificate

Actor: Student / System

Precondition: Student must satisfy the course completion requirements.

Main Flow:

1. System checks the student's course completion status.
2. System verifies assessment requirements.
3. If the student is eligible, the system generates a certificate.
4. Certificate is stored in the student's account.
5. Student can view or download the certificate.

Postcondition: Course completion certificate is available to the student.

4.7 Data Requirements

The system shall maintain the following major data:

User Data

- User ID
- Name
- Email
- Password
- Role
- Contact information

Course Data

- Course ID
- Course title
- Description
- Category
- Instructor ID
- Course duration
- Course status

Learning Material Data

- Material ID
- Course ID

- Material title
- Material type
- File location
- Upload date

Enrollment Data

- Enrollment ID
- Student ID
- Course ID
- Enrollment date
- Completion status

Assessment Data

- Assessment ID
- Course ID
- Questions
- Answers
- Student score
- Assessment date

Progress Data

- Progress ID
- Student ID
- Course ID
- Completed lessons
- Completed assessments
- Completion percentage

Certificate Data

- Certificate ID
- Student ID
- Course ID
- Issue date
- Certificate status

4.8 External Interface Requirements

User Interface

The system should provide:

- Login page
- Registration page
- Student dashboard
- Instructor dashboard
- Administrator dashboard

- Course listing page
- Course details page
- Learning material page
- Assessment page
- Progress dashboard
- Certificate page

Hardware Interface

The system should be accessible using:

- Desktop computers
- Laptops
- Tablets
- Smartphones

Software Interface

The system may interact with:

- Web browsers
- Database management system
- File storage system
- Email/notification services

Communication Interface

The system shall use secure internet communication such as **HTTPS** for transmitting user and system data.

4.9 System Features

Feature 1 – User Management

Allows users to register, log in, and manage their profiles.

Feature 2 – Course Management

Allows instructors to create, update, publish, and manage courses.

Feature 3 – Learning Material Management

Allows instructors to upload and organize educational resources.

Feature 4 – Online Learning

Allows students to access course materials through the platform.

Feature 5 – Assessment System

Allows instructors to create assessments and students to attend them online.

Feature 6 – Progress Tracking

Displays the student's learning progress and course completion percentage.

Feature 7 – Performance Analysis

Stores and displays assessment scores and learning performance.

Feature 8 – Certificate Generation

Automatically generates certificates for students who successfully complete the course requirements.

Feature 9 – Notification System

Provides notifications about courses, assessments, results, and certificates.

5. Assumptions and Constraints

5.1 Assumptions

1. Users have a valid internet connection.
2. Students and instructors provide correct registration information.
3. Instructors upload legally permitted educational content.
4. Students use supported web browsers.
5. The system database is available for storing user and course information.
6. Students complete required assessments within the specified time.
7. Certificate eligibility criteria are defined by the instructor or administrator.

5.2 Constraints

Technical Constraints

- The system requires internet connectivity.
- Server and database resources may limit the number of simultaneous users.
- Large video and document files require sufficient storage capacity.

Operational Constraints

- Users must have valid login credentials.
- Instructors are responsible for maintaining accurate course content.
- Students must complete required learning activities to receive certificates.

Economic Constraints

- Hosting and cloud storage may involve operational costs.
- Maintenance and server resources may require additional expenses.

Legal Constraints

- User data must be protected.
- Copyrighted learning materials must be used with proper authorization.
- The system should follow applicable data protection and privacy requirements.

Time Constraints

- The system should be developed within the project schedule.
- Major features should be implemented and tested before final deployment.

6. Validation of Requirement Completeness and Consistency

The requirements are validated based on the following criteria:

Completeness

All major stakeholder requirements have been addressed, including:

- Instructor course management.
- Learning material upload.
- Student enrollment.
- Online assessments.
- Progress tracking.
- Performance monitoring.
- Certificate generation.

Consistency

The requirements do not conflict with each other. Instructor-related requirements focus on course and assessment management, while student-related requirements focus on learning and progress.

Ambiguity

Requirements are defined using clear terms such as course enrollment, assessment, progress tracking, and certificate generation.

Redundancy

Duplicate requirements have been avoided by grouping related functions under appropriate modules.

Missing Requirements

Important supporting requirements such as authentication, notification, data storage, security, and administrator management have also been included.

Suggested Improvements

The system can be enhanced in the future by adding:

- Live online classes.
- Discussion forums.
- AI-based course recommendations.
- Personalized learning paths.
- Gamification and badges.
- Mobile application support.
- Advanced learning analytics.
- Automated plagiarism detection.

Conclusion

The **Online Course Learning Management System** provides a centralized platform for online education. It enables instructors to create courses, upload learning materials, and conduct assessments, while students can enroll in courses, access learning resources, attend assessments, track their progress, and receive certificates.

The proposed requirements cover the major functional and non-functional needs of the system and provide a clear foundation for designing and developing a scalable and user-friendly e-learning platform.